Problem 1

PF of $12 = 2 \times 6$

$= 2 \times 2 \times 3$

$= 2^2 \times 3^1$

PF of $18 = 2 \times 9$

$= 2 \times 3 \times 3$

$= 2^1 \times 3^2$

$GCD(12, 18) = 2^1 \times 3^1$

$= 2 \times 3$

$= 6$

No. of 2's powers $= 1 + 1$

$= 2$

No. of 3's powers $= 1 + 1$

$= 2$

It's confusing

Total factors $= 2 \times 2$

$= 4$

$= (1+1)(1+1) = 4$

List of factors = $[(2^0 \times 3^0), (2^1 \times 3^0), (2^0 \times 3^1), (2^1 \times 3^1)]$

$= [1, 2, 3, 6]$

Problem 2:

PF of 24 $= 8 \times 3$

$= 2 \times 2 \times 2 \times 3$

$= 2^3 \times 3^1$

PF of 36 $= 2^2 \times 3^2$

GCD (24, 36) $= 2^2 \times 3^1$

$= 4 \times 3$

$= 12$

No. of 2's powers $= 2 + 1$

$= 3$

No. of 3's powers $= 1 + 1$

$= 2$

Total no. of factors $= 3 \times 2$

$= 6$

List of factors $= [(2^0 \times 3^0), (2^1 \times 3^0), (2^2 \times 3^0), (2^0 \times 3^1), (2^1 \times 3^1), (2^2 \times 3^1)]$

$= [1, 2, 4, 3, 6, 12]$

Problem 3:

PF of 40 $= 8 \times 5$

$= 2 \times 2 \times 2 \times 5$

$= 2^3 \times 5^1$

PF of 60 $= 12 \times 5$

$= 2 \times 2 \times 3 \times 5$

$= 2^2 \times 3^1 \times 5^1$

GCD $= 2^2 \times 3^0 \times 5^1 \quad = 2^2 \times 5^1$

$= 4 \times 5$

$= 20$

No. of 2's powers $= 2 + 1$

$= 3$

No. of 3's powers $= 0 + 1$

$= 1$

No. of 5's powers $= 1 + 1$

$= 2$

Ans $= (2+1)(1+1)$

$= 3 \times 2 = 6$

Total factors $= 3 \times 1 \times 2$

$= 6$

List of factors =

$= [(2^0 \times 3^0 \times 5^0), (2^1 \times 3^0 \times 5^0), (2^2 \times 3^0 \times 5^0),$

$(2^0 \times 3^0 \times 5^1), (2^1 \times 3^0 \times 5^1), (2^2 \times 3^0 \times 5^1)]$

$= [1, 2, 4, 5, 10, 20]$

✓

Problem 4:

PF of 15 $= 5 \times 3$

$= 5^1 \times 3^1$

PF of 75 $= 25 \times 3$

$= 5 \times 5 \times 3$

$= 5^2 \times 3^1$

GCD $= 5^1 \times 3^1$

$= 15$

No. of 5's powers $= 1 + 1$

$= 2$

No. of 3's powers $= 1 + 1$

$= 2$

$2 \times 2 = 4$

List of factors $= [(5^0 \times 3^0), (5^1 \times 3^0), (5^0 \times 3^1), (5^1 \times 3^1)]$

$= [1, 5, 3, 15]$ ✓

Problem 5:

$$PF(48) = 16 \times 3$$
$$= 2 \times 2 \times 2 \times 2 \times 3$$
$$= 2^4 \times 3^1$$

$$PF(80) = 8 \times 10$$
$$= 2 \times 2 \times 2 \times 2 \times 5$$
$$= 2^4 \times 5^1$$

$$GCD = 2^4 \times 3^0 \times 5^0$$
$$= 16$$

No. of 2's powers $= 4 + 1$
$= 5$

$$\left.\begin{array}{l}\text{No. of 3's powers} = 0 + 1 \\ = 1 \\ \text{No. of 5's powers} = 0 + 1 \\ = 1\end{array}\right\} \text{Not needed}$$

$$5 \times \underbrace{1 \times 1}_{\text{Not needed}} = 5$$

List of factors $= [(2^0 \times 3^0 \times 5^0), (2^1 \times 3^0 \times 5^0), (2^2 \times 3^0 \times 5^0), (2^3 \times 3^0 \times 5^0), (2^4 \times 3^0 \times 5^0)]$

$= [0, 2, 4, 8, 16]$

1

Problem 6:

$PF(100) = 4 \times 25$

$= 2 \times 2 \times 5 \times 5$

$= 2^2 \times 5^2$

$PF(150) = 50 \times 3$

$= 2 \times 5 \times 5 \times 3$

$= 2^1 \times 5^2 \times 3^1$

$GCD = 2^1 \times 5^2 \times 3^0$

$= 50$

No. of 2's powers $= 1 + 1$

$= 2$

No. of 5's powers $= 2 + 1$

$= 3$

No. of 3's powers $= 0 + 1$

$= 1$

$2 \times 3 \times 1 = 6$ ✓

List of factors = [($5^0 \times 2^0 \times 3^0$), ($5^1 \times 2^0 \times 3^0$), ($5^2 \times 2^0 \times 3^0$), ($5^1 \times 2^1 \times 3^0$), ($5^2 \times 2^1 \times 3^0$)]

($5^0 \times 2^1$)

= [1, 5, 25, 10, 50]

2

$\cancel{2}$

Problem 7:

$PF(56) = 8 \times 7$

$= 2 \times 2 \times 2 \times 7$

$= 2^3 \times 7^1$

$PF(98) = 14 \times 7$

$= 2 \times 7 \times 7$

$= 2^1 \times 7^2$

$GCD = 2^1 \times 7^1$

No. of 2's factors $= 1 + 1$

$= 2$

No. of 7's factors $= 1 + 1$

$= 2$

$2 \times 2 = 4$

List of factors $= [(2^0 \times 7^0), (2^1 \times 7^0), (2^0 \times 7^1), (2^1 \times 7^1)]$

$= [1, 2, 7, 14]$

Problem 8:

$PF(81) = 3 \times 27$

$= 3 \times 3 \times 3 \times 3$

$= 3^4$

$PF(27) = 3 \times 9$

$= 3 \times 3 \times 3$

$= 3^3$

$GCD = 3^3$

$= 27$

No. of 3's powers $= 3 + 1$

$= 4$

List of factors $= [3^0, 3^1, 3^2, 3^3]$

$= [1, 3, 9, 27]$ ✓

Problem 9:

$PF(45) = 5 \times 9$

$= 5 \times 3 \times 3$

$= 5^1 \times 3^2$

$PF(105) = 5 \times 21$

$= 5 \times 7 \times 3$

$= 5^1 \times 7^1 \times 3^1$

$GCD = 5^1 \times 7^0 \times 3^1$

$= 15$

No. of 5's factors $= 1 + 1$

$= 2$

No. of 7's factors $= 0 + 1$

$= 1$

No. of 3's factors $= 1 + 1$

$= 2$

$2 \times 1 \times 2 = 4$ ✓

List of factors = [($5^0 \times 7^0 \times 3^0$),

($5^1 \times 7^0 \times 3^0$),

($5^0 \times 7^0 \times 3^1$),

($5^1 \times 7^0 \times 3^1$)]

= [1, 5, 3, 15]

Problem 10:

$$PF(120) = 2 \times 60$$
$$= 5 \times 3 \times 2 \times 2 \times 2$$
$$= 5^1 \times 3^1 \times 2^3$$

$$PF(160) = 8 \times 20$$
$$= 2 \times 2 \times 2 \times 2 \times 2 \times 5$$
$$= 2^5 \times 5^1$$

$$GCD = 2^3 \times 5^1 \times 3^0$$
$$= 40$$

No. of 2's powers $= 3 + 1$
$= 4$

No. of 5's powers $= 1 + 1$
$= 2$

No. of 3's powers $= 0 + 1$
$= 1$

$4 \times 2 \times 1 = 8$ ✓

List of factors = [($2^0 \times 5^0 \times 3^0$),
($2^1 \times 5^0 \times 3^0$),
($2^2 \times 5^0 \times 3^0$),
($2^3 \times 5^0 \times 3^0$),
($2^0 \times 5^1 \times 3^0$),
($2^1 \times 5^1 \times 3^0$),
($2^2 \times 5^1 \times 3^0$),
($2^3 \times 5^1 \times 3^0$)]

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1, 2, 4, 5, 8, 10, 20, 40